Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 26)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
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9	
10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 . 1 Explain how affluence and the strategies of manufacturers increase the amount of solid waste produced.

[4 marks]

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0 1 . 2 Give **three** factors that affect the choice of disposal method for a solid waste.

[3 marks]

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0 1 . 3 Explain **one** environmental advantage and **one** disadvantage of composting garden and food waste rather than sending it to landfill.

[2 marks]

Advantage .....

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Disadvantage .....

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0 1 . 4 Explain why mixed municipal waste is difficult to recycle.

[1 mark]

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0 2 . 1 Describe how an **electrostatic precipitator** removes smoke particles from the effluent gases of a coal-fired power station.

[3 marks]

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0 2 . 2 Explain how a **cyclone separator** works.

[2 marks]

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0 2 . 3 Explain why heating coal to remove tar reduces smoke emissions.

[1 mark]

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0 2 . 4 Explain why turbochargers reduce the smoke produced by diesel engines.

[2 marks]

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0 2 . 5 Explain why smaller smoke particles (PM1) are a greater health risk than larger ones.

[2 marks]

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0 3 . 1 Which of the following water sources is usually the most convenient for large communities? Tick one box.

[1 mark]

A Desalinated seawater ☐

B River water ☐

C Rainwater catchment ☐

D Newly discovered aquifers ☐

0 3 . 2 Explain why the quality requirements for water differ between uses, using **two** named uses.

[2 marks]

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0 3 . 3 Explain why inter-basin transfers may be used and give **one** disadvantage.

[2 marks]

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0 4 . 1 Which of the following properties would make a pesticide **least** likely to biomagnify? Tick one box.

[1 mark]

A High liposolubility and high persistence ☐

B Low liposolubility and low persistence ☐

C High liposolubility and low persistence ☐

D Low liposolubility and high persistence ☐

0 4 . 2 Explain the difference between **bioaccumulation** and **biomagnification**.

[2 marks]

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0 4 . 3 Explain why pesticides are described as an unusual type of pollutant.

[2 marks]

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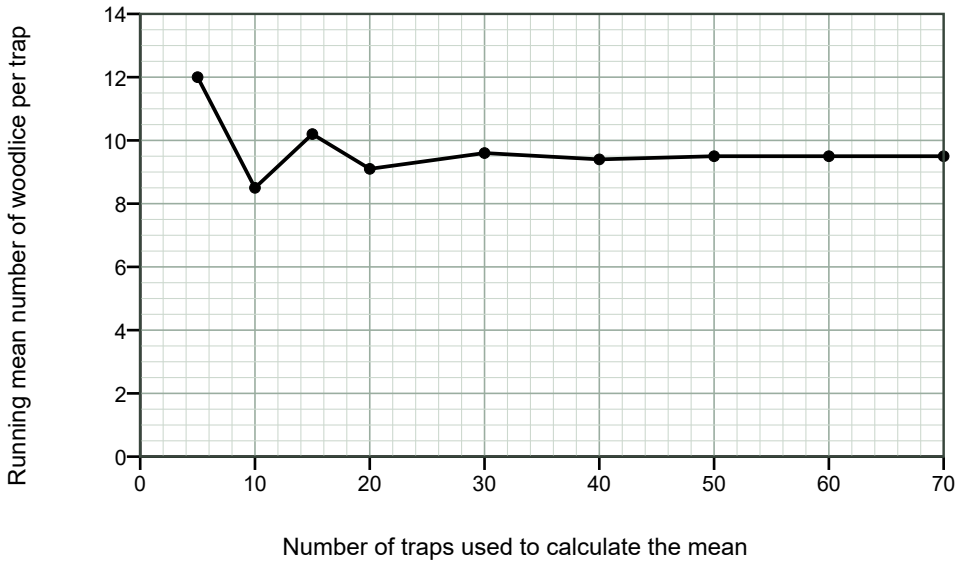
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0 5 Figure 1 shows the mean number of woodlice per pitfall trap calculated as more traps were included in the mean (a running mean).

Figure 1



0 5 . 1 Use Figure 1 to suggest the minimum number of traps needed to produce a reliable mean. Give a reason for your answer.

[2 marks]

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0 5 . 2 Explain why the number of samples needed depends on the degree of scatter of the results.

[2 marks]

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0 5 . 3 The student wants to compare woodlouse numbers in a woodland and in a grassland. Describe how the sampling should be standardised.

[3 marks]

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0 5 . 4 Explain why the timing of sampling would matter if the student were investigating the activity of flying insects rather than woodlice.

[3 marks]

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0 6 . 1 Explain why mineral extraction causes more land-use conflict than most other human activities.

[2 marks]

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0 6 . 2 Describe how each of the following environmental impacts of an operating mine can be reduced.

[5 marks]

Dust .....

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Noise .....

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Turbid drainage water .....

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Acidic leachate from spoil .....

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Loss of amenity .....

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0 6 . 3 Explain how spoil heaps can be made stable and describe what happened at Aberfan in 1966.

[3 marks]

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0 7 . 1 Explain how climate change may affect the survival of species. Use named examples.

[6 marks]

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0 7 . 2 Explain why a species living in an interdependent community may be unable to move to a new area even if suitable conditions exist there.

[2 marks]

0 7 . 3 Explain why polar regions warming faster than the equator affects the weather in the UK.

[2 marks]

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0 8 A nuclear fuel reprocessing plant discharges small amounts of radioactive material in liquid effluent to a coastal sea.

0 8 . 1 Explain why it cannot be assumed that the discharge will be diluted to insignificant levels.

[2 marks]

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0 8 . 2 Give **four** factors that would be included in a Critical Pathway Analysis of this discharge.

[4 marks]

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0 8 . 3 Explain how the results of the analysis reduce the cost of environmental monitoring.

[2 marks]

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0 8 . 4 Explain why Critical Group Monitoring measures exposure rather than damage to health.

[2 marks]

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0 9 . 1 Explain the difference between nuclear fission and nuclear fusion.

[2 marks]

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0 9 . 2 State the sources of the two fuels used in fusion research.

[2 marks]

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2 .....

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0 9 . 3 Explain why very high temperatures, a vacuum and a magnetic field are needed in a toroidal (torus) fusion reactor.

[3 marks]

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0 9 . 4 Give **two** potential advantages of fusion over fission and one reason why it is not yet a commercial energy source.

[3 marks]

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1 0 Table 1 shows the three forms of oxygen found in the stratosphere.

Table 1

Form of oxygen	Number of atoms per molecule
Monatomic oxygen (O)	1
Diatomic oxygen (O_2)	2
Triatomic oxygen / ozone (O_3)	3

1 0 . 1 Explain how the three forms of oxygen protect life at the surface from ultraviolet radiation.

[3 marks]

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1 0 . 2 Give **two** other ways in which the atmosphere supports life.

[2 marks]

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2 .....

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1 0 . 3 Explain why the atmosphere is thermally stratified into layers.

[1 mark]

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1 0 . 4 The troposphere and stratosphere are the layers most affected by human activities. Discuss how human activities alter the energy processes in these two layers and evaluate the difficulties of predicting the consequences.

[9 marks]

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Extra space

15

Write an essay on one of the following topics.

1 1 . 1 Evaluate the contribution of new technologies to the monitoring and understanding of environmental change in the physical environment.

[25 marks]

OR

1 1 . 2 Mineral resources are non-renewable. Discuss how an understanding of geological processes and the development of new technologies can extend the period for which society can rely on mineral resources.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS